

# Physics of Cognition: Homework 3

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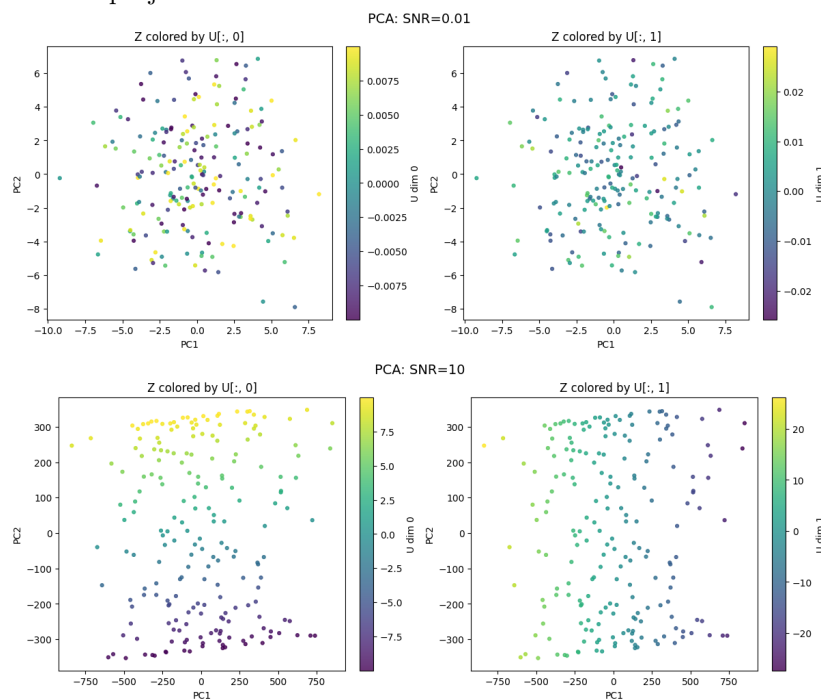
February 15th, 2026

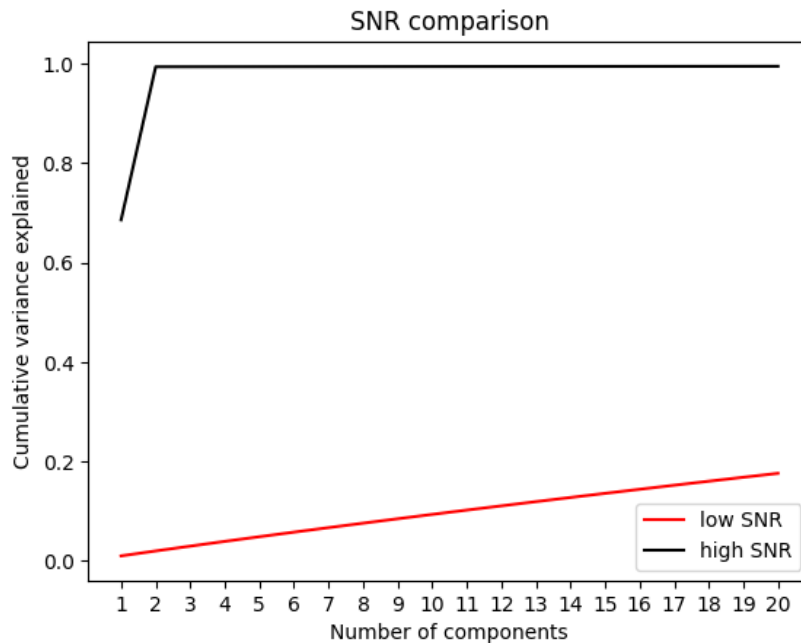
## 1.1: PCA Intro

1. This random  $V_x$  is merely transforming some space into a different dimension while preserving the Euclidean distances of each point since the projection is fully orthogonal to the starting space.

## 1.2: PCA analysis

1.  $W$  is the projection matrix. Size should be  $d \times 2$ .



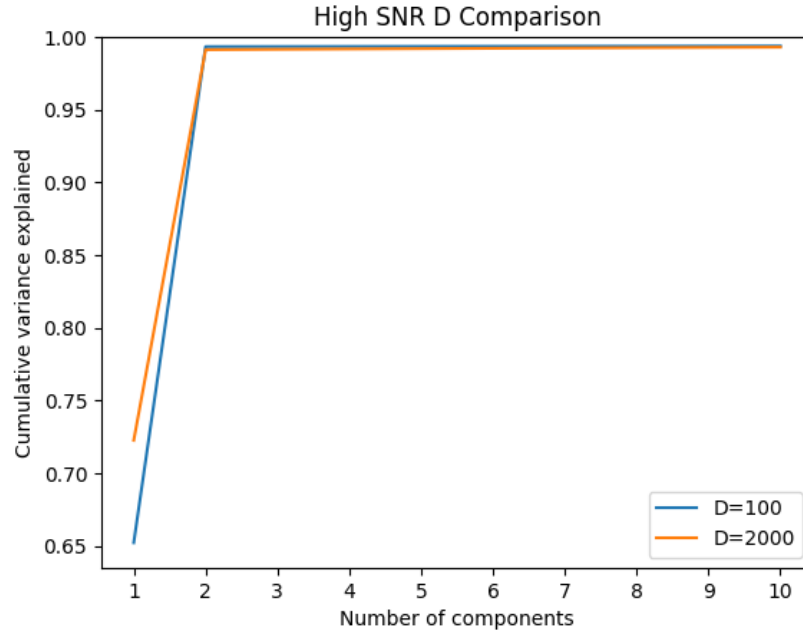


4. PCR in the low SNR dataset can barely explain the variance in that distribution while the high SNR plot shows that we quickly converge to almost explaining the full variance. If we include more components, the low SNR plot could steadily rise, but it shows how PCR is struggling there.

5. Even though  $X^T X$  isn't invertible, we aren't exactly looking for invertibility since we can still compute the eigenvalues and get principal components from these quantities.

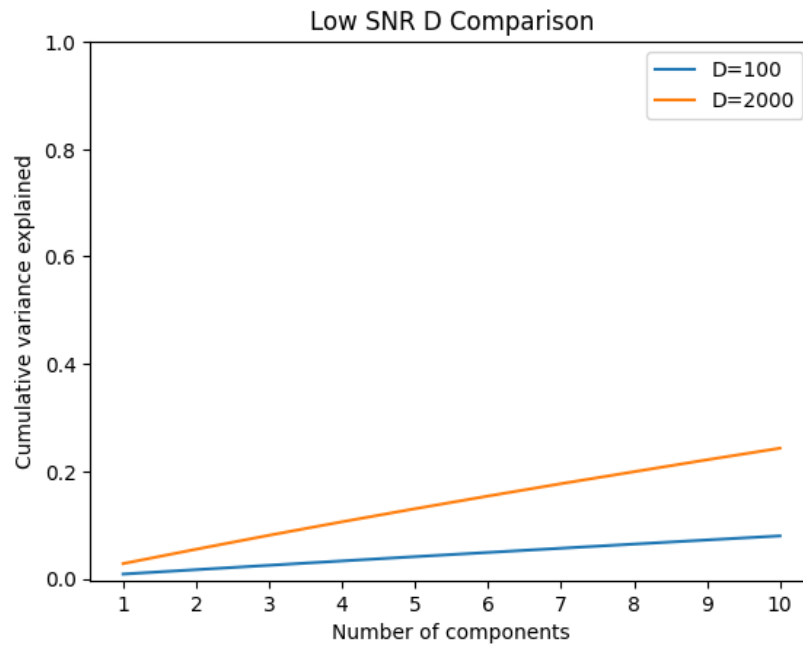
### 1.3: Let's play around with PCA parameters

When  $D$  varies between 100 and 2000, there isn't much of a difference in how much each PCA component captures variance.

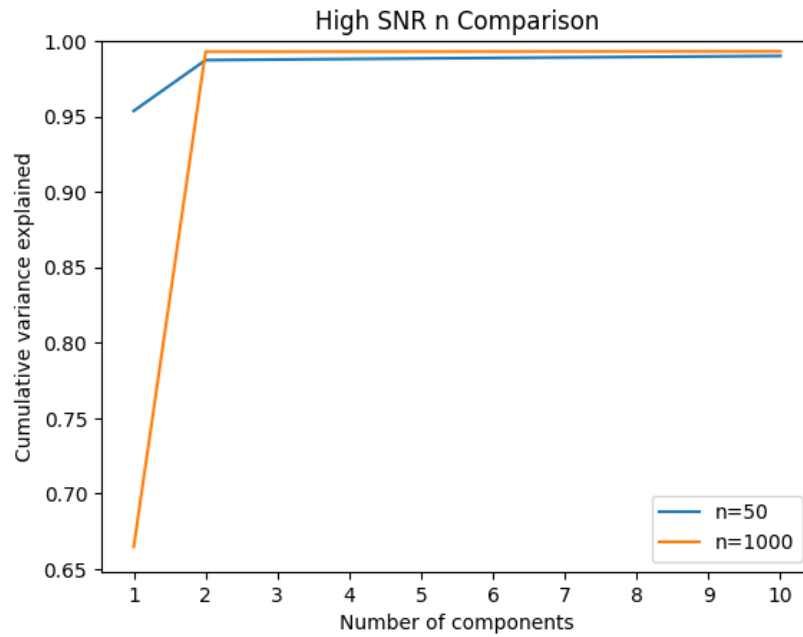


This is likely since all the additional dimensions with the latter setup mostly collapse to the same principal components since the eigenvalues are well differentiated.

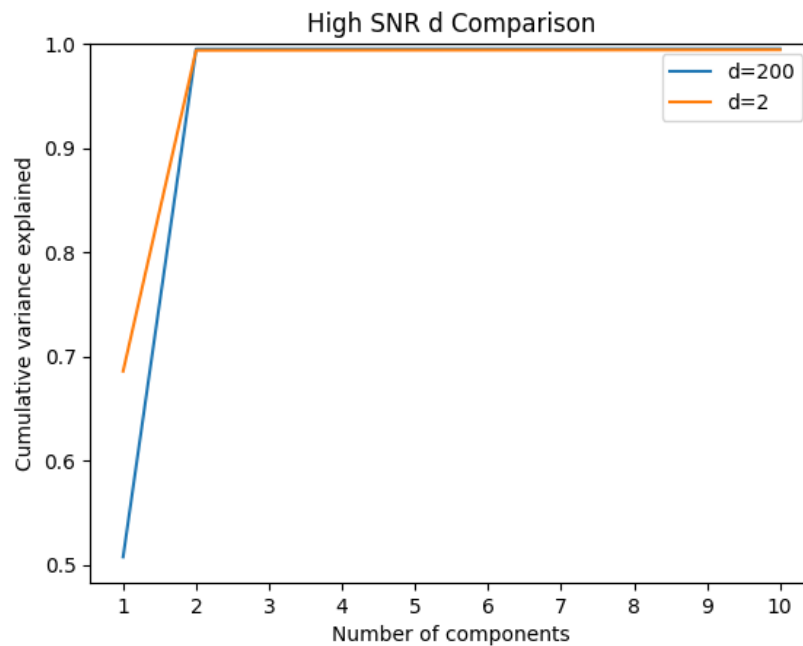
However, repeating the experiment with low SNR, we see that there's a much clearer difference in cumulative variance. There is no more signal in  $D = 2000$  vs  $D = 100$ , but there's the chance for some random correlation between the noise and signal which boosts the cumulative curve. I'm not entirely sure why this is.



When increasing the sample size, we see a large jump with more sample sizes since the initial eigenvalues likely can't capture as much variance initially with more data points but it quickly catches up after we reach a sufficient number of components such that there's only a slight difference in the ultimate cumulative variance.



Finally, we look at changing the dimension of the signal itself (intrinsic dimension).



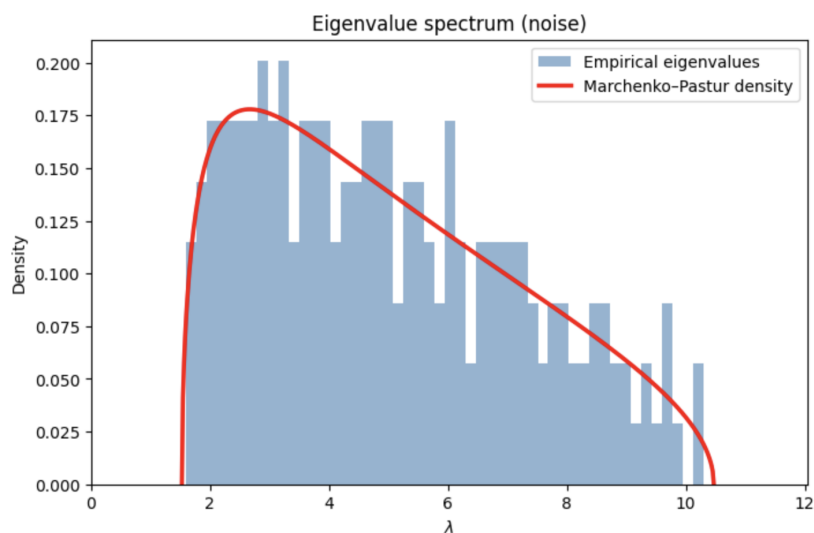
We see that with a lower signal dimension we are getting a higher variance explained since there's less signal to capture. However, within two components,

we see the graphs converging.

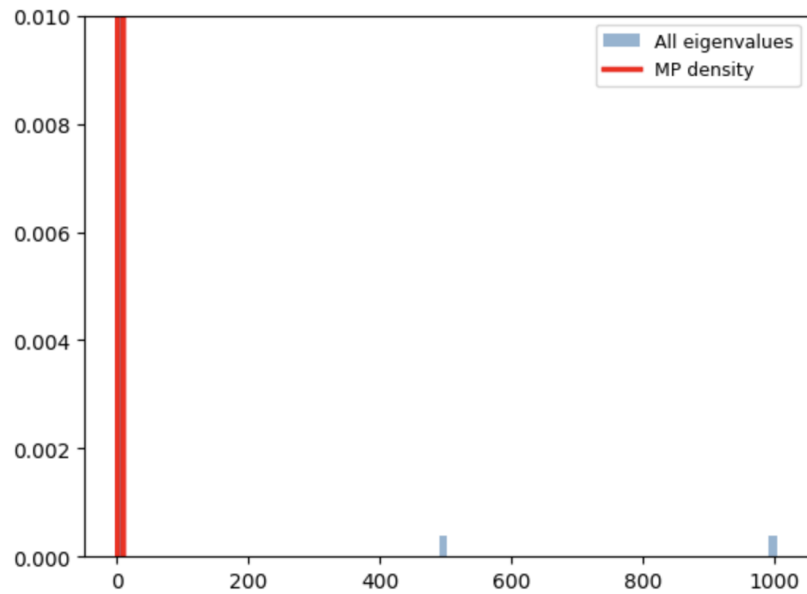
## 1.4: Discussion

1. The fundamental idea behind vanilla PCA is that a ranking of the variances can be determined from the eigenvalues of the covariance matrix. If the actual variance of the signal cannot be described in a linear way, PCA will be inefficient at determining components that explain the variance (likely overestimating). Also, if the true factors underlying the data are correlated, PCA may perform poorly since each component is constrained to be orthogonal.
2. There are cases—such as studies recording neurons while monkeys reach—where the variance in the neural population is rotational, which traditional PCA does not capture well. Additionally, PCA has no explicit notion of time; it may split the data into multiple dimensions to approximate dynamics that are actually generated by a simpler underlying temporal process.

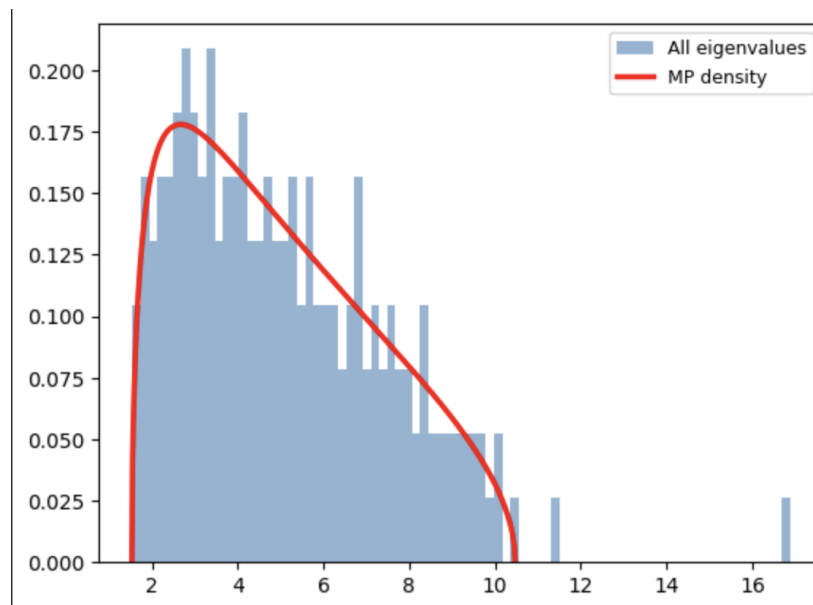
## 2.1: Just noise



## 2.2: Adding some signal



Yes, adding signal actually produces eigenvalues far outside the distribution! You could barely see the rest of the eigenvalues on the left.



When using a much lower SNR ( $=0.1$ ), there are still outliers but their

eigenvalues are much closer to the bulk.

## 2.3: Discussion

1. When an eigenvalue is penetrating the MP distribution on the horizontal axis (meaning a larger eigenvalue than it captures, it's actually capturing signal as we saw in the plots.
2. It's likely that certain bits of random noise are related to each other so the covariance matrix picks this up as possible directions of variance and creates a distribution.
3. If not scrutinized enough, data could appear to have few correlations despite having large eigenvalues and signal embedded within, which is essential when looking at activity from neurons and understanding trends across the components or in finance where slight nuanced correlations are vital to the market as a whole. To combat this, using methods like the Marchenk-Pastur distribution to assign a safe noise range for the eigenvalues is greatly helpful.

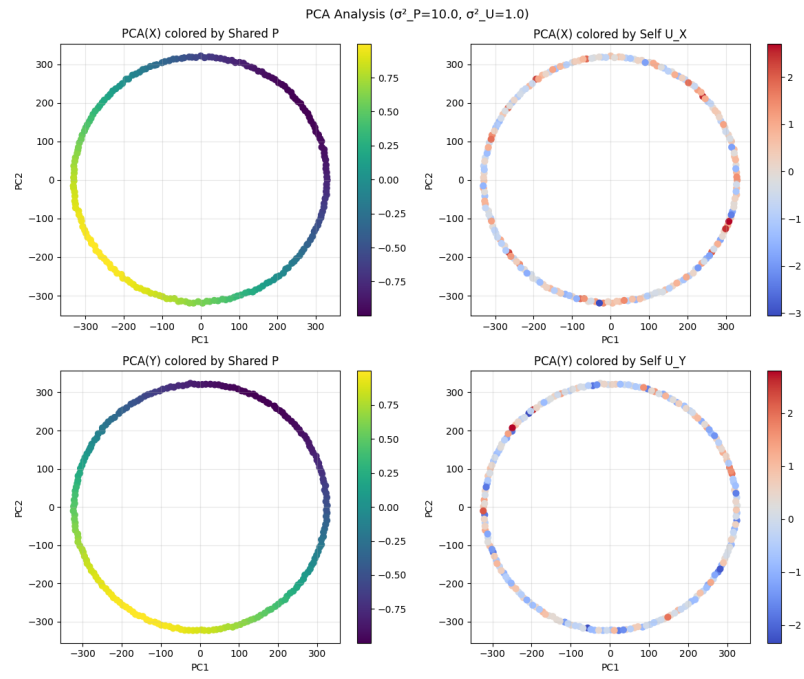
## 3.1 Raw Signal Visualization

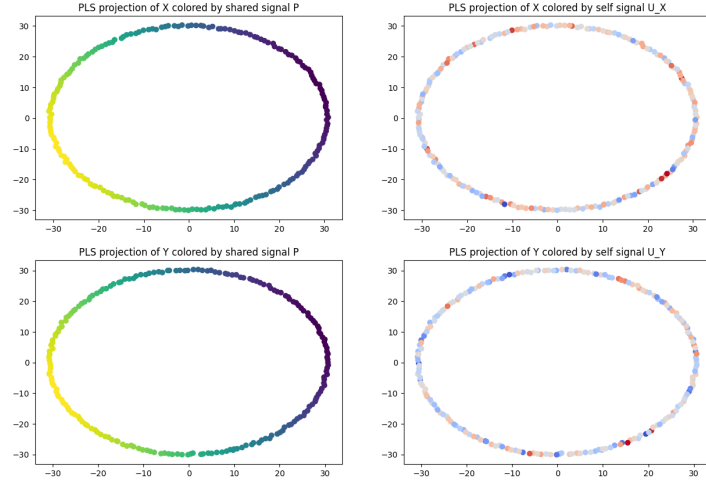




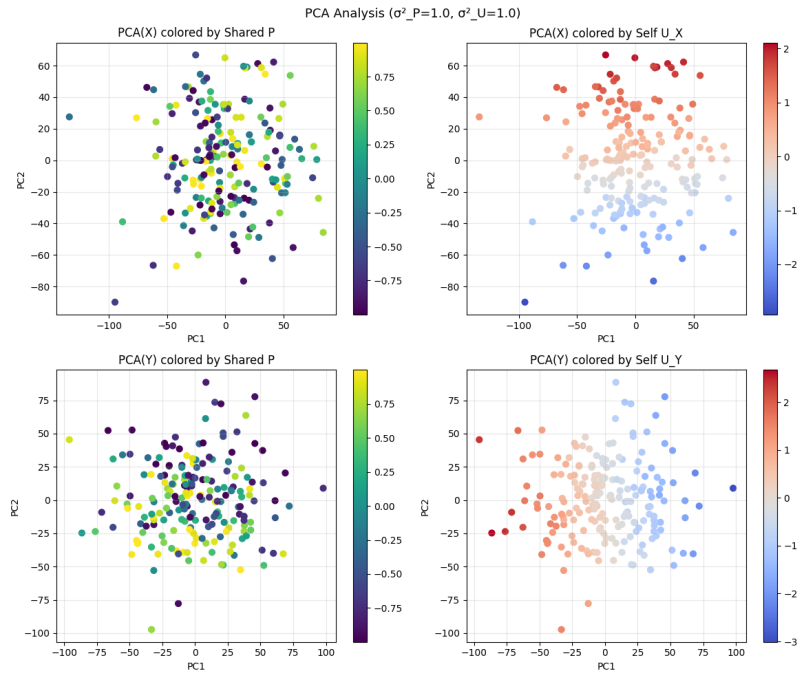
## 3.2 PLS vs PCA Analysis

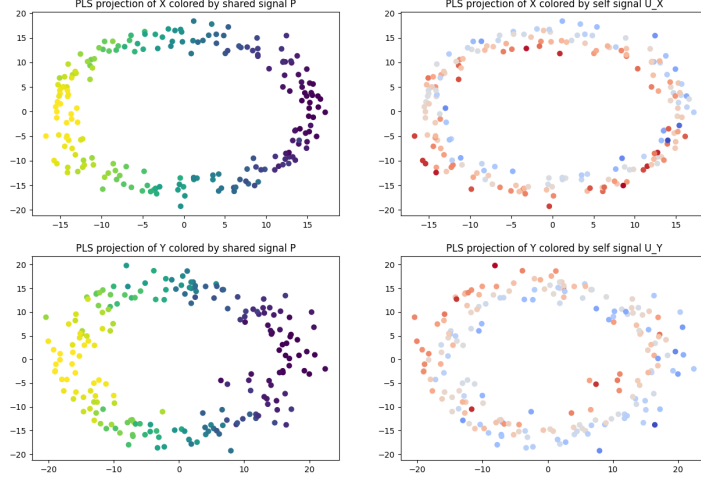
High Shared SNR ( $\sigma_P^2 = 10$ )





Low Shared SNR ( $\sigma_P^2 = 1$ )





When the SNR is high, the PCR picks up the signal for X and Y, and the PLS plot is the exact same for X and Y when colored by the shared signal P (proving it works!) while the self  $U_x/U_y$  graphs are messy since there isn't anything to pick up on.

But, of course, with a more diluted signal, the shape of the principle components are less defined and better represented by the noise (as shown in the right plots). However, with PLS, we still capture the majority of the signal  $P$  even though the signal is too sparse for PCR.

### 3.3: Discussion

1. We can see this case better in the lower SNR regime. This is because PCA looks at the individual covariances of each dataset without regard for any similarity with another. Since  $U_X$  has more variance than  $P$ , the right side graphs look a much nicer with PCA. However, PLS looks for shared structure among X and Y even if this structure ( $P$ ) has lower variance. This is why the plots on the left of the PLS projection are a smooth transition of colors.
2. PLS works well with low SNR datasets because it isn't only incentivized to find the maximal variance within each dataset but between the two. The independent noises and self-signals ( $R, U$ ) have close to no covariance between X, Y while  $P$  is the only source of the correlation.
3. We might use PCA when we're looking at single datasets and know that the variances exist within. But PLS is favorable when there exists a com-

mon signal between two datasets which might not have high variance independently but cause the most correlation between the two.

4. Sometimes we might be looking at neuron activity from vision and see hints of medium-variance principle components we can extract, not realizing that there's an even lower-variance signal existing in another set of neurons responsible for hearing that PCA completely misses.

## 4: Reflection

1. The naive approach to understand neuron behavior is to bunch together signals from the sample and plot covariances between each pair. This would be very slow and tedious since there's so much noise and variability with each signal, which is where PCA comes in. We can extract principle components using the magic of linear algebra to see directions of highest variance. Now, we can model entire populations of neurons and extract information about their collective patterns. This was the point of investigation in Churchland et al. where it was discovered that rotational dynamics were at play during reaching which fundamentally changed our understanding of how motor neurons encode movement parameters. jPCA allowed this team to see rotations which typical linear PCA couldn't cleanly represent. However, there's the risk that random noisy fluctuations are seen as signal depending on the under/over-parameterization of the data ( $n > D$  or  $n < D$ ) in problems. Many of the techniques we've been learning are based on the low-rank signal paradigm where there are latent factors controlling the neuron behaviors—PCA, PLS, etc make sense here since they explicitly break down the data into more abstract dimensions. The alternate perspective is there is signal within every single neuron that could be treated as nodes in a graph, each having a role in the overall goal.
2. This homework took a while, close to 7 hours. Much of it was tinkering with the plotting and seeing whether I was getting the results I was supposed to. The exercise looking at the Marchenko–Pastur Distribution was very fascinating because I could see how adding signal (and tampering with the SNR) changes the magnitude of the largest eigenvalues. Also, the PLS exercises were laid out nicely to show how this new method uncovered structure from the low SNR data that PCR couldn't do. All in all, tough but very enjoyable and thought-provoking.

Colab notebook